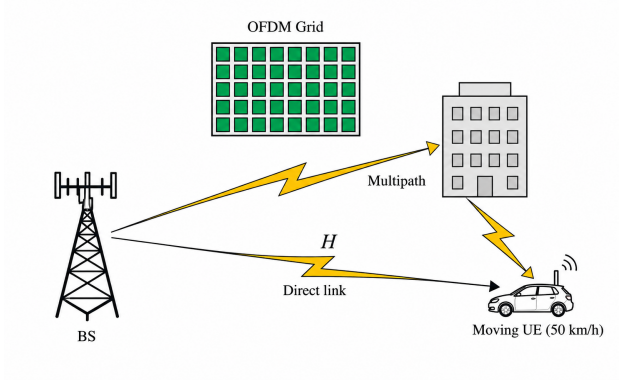
Deep Learning Based Channel Estimation for 5G/6G Wireless Network

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MTP Mid-Semester Presentation

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Why Channel Estimation Matters



Base station to moving receiver through a multipath channel

Why is it needed?

- Multipath reflections cause signal attenuation and phase shifts.
- Receiver motion introduces time-varying channel effects.
- Each OFDM subcarrier experiences a different channel gain.
- Channel estimation enables equalization and reliable symbol detection.

Our Motivation

Estimate the full channel from sparse pilot observations while reducing pilot overhead.

Background: LS and LMMSE Channel Estimation

Pilot-assisted estimation inserts known symbols; the receiver infers the full channel from noisy, sparse pilot observations

$$Y_p = H_p X_p + W_p.$$

Least Squares (LS)

$$\widehat{H}_{\mathrm{LS}} = \frac{Y_p}{X_p}$$

- Simple and does not require channel statistics.
- Noise-sensitive; only known at pilot locations.
- Requires interpolation/extrapolation over the grid.

Linear MMSE (LMMSE)

$$\widehat{h}_{\mathrm{LMMSE}} = A\widehat{h}_{\mathrm{LS}}$$

- Uses channel covariance and noise variance.
- Usually improves MSE over LS.
- Requires prior statistics and matrix computation.

Literature Review

- Traditional estimators: LS, MMSE, and LMMSE balance complexity, pilot overhead, and statistical knowledge.
- CNN denoisers and super-resolution approaches treat the channel grid as a two-dimensional image.
- ChannelNet, ReEsNet, and RRDBNet use convolutional/residual learning for channel recovery.
- Attention mechanisms recalibrate feature channels, potentially suppressing noise-related features.
- The base paper combines residual learning with channel attention in AttRNet-Conv and AttRNet-FC.

See references [1]–[6].

MTP Aims and Objectives

1. Reproduce and validate the base paper: OFDM system model, LS/LMMSE estimators, AttRNet-Conv, and AttRNet-FC.
2. Study reduced pilot overhead through pilot-count and pilot-placement configurations.
3. Design a residual Transformer/self-attention estimator and benchmark it against AttRNet.
4. Compare NMSE/MSE, SNR response, pilot overhead, and computational complexity.
5. Extend the resulting framework to IRS-assisted OFDM with cascaded-channel estimation and IRS phase configuration.

Base Paper: AttRNet

- Input: real and imaginary LS pilot estimates.
- Stem 3×3 convolution followed by three attention-residual blocks.
- Each block: three 3×3 convolutions, global-average pooling, excitation, feature scaling, and residual addition.
- Bilinear upsampling reconstructs the full 72×14 grid.
- FC attention: two fully connected excitation layers.
- Conv attention: two 1×1 convolutional excitation layers.

Implemented Check

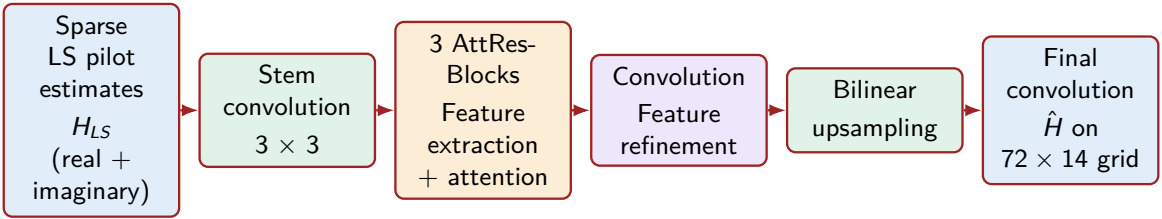
8 feature channels

3 residual-attention blocks

9,670 trainable parameters

This parameter count matches Table II of the base paper.
The final 36×7 convolution uses same-size asymmetric padding because the paper does not specify its padding/stride.

AttRNet: Learning-Based Channel Reconstruction



What the network learns

- Convolutions learn local time-frequency patterns from sparse LS estimates.
- Attention weights recalibrate learned feature maps.
- Upsampling and the final convolution reconstruct the full channel grid.

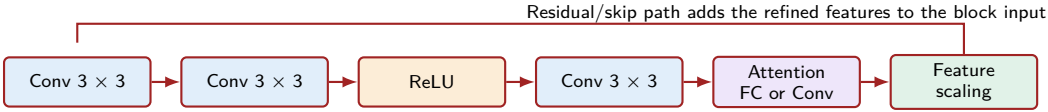
Training objective

The predicted complex channel is compared with the simulated ground truth using MSE:

$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^{N} \left| H_i - \hat{H}_i \right|^2 .$$

Gradients update convolution and attention parameters end-to-end.

Inside an AttResBlock: FC vs Conv Attention



AttRNet-FC

Global average pooling summarizes each feature map. Two fully connected layers and a sigmoid produce one weight per learned feature channel.

AttRNet-Conv

The excitation stage uses two 1×1 convolutions, ReLU and sigmoid to produce feature-channel weights.

Attention weights refer to learned CNN feature channels, not directly to individual pilot positions.

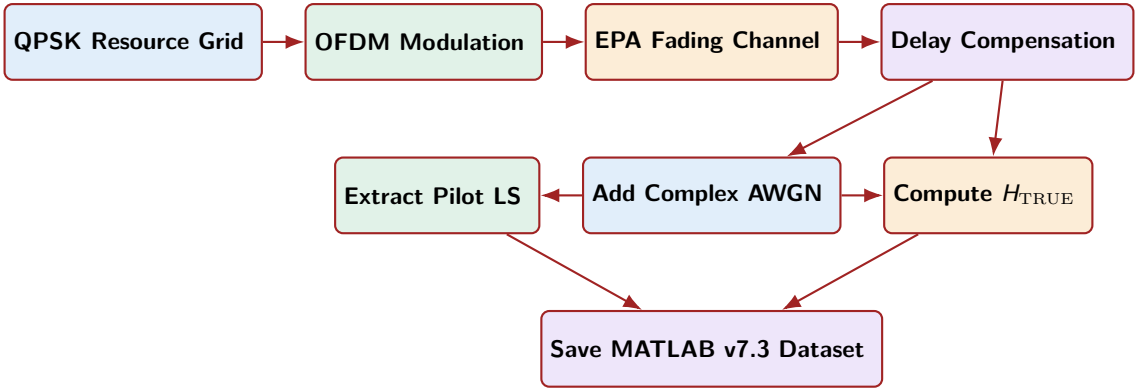
Dataset Generation

Simulation Setup

Channel	MATLAB `lteFadingChannel`; 3GPP EPA, 7 paths; GMEDS fading
Carrier / motion	2.5 GHz / 50 km/h ($f_D \approx 115.82$ Hz)
OFDM	72 subcarriers, 14 symbols, 15 kHz spacing, QPSK, CP=16
Pilots	24: 6×4 grid; 48: 12×4 grid
Data split	20,000 samples: 12,000 train / 4,000 validation / 4,000 test
SNRs	0, 5, 10, 15, 20 dB; equal allocation
Optimizer	Adam, learning rate 10^{-3}, batch size 128, 100 epochs
Evaluation	Held-out MATLAB test set; MSE averaged separately at each SNR

Important:
Uniform pilot coordinates and equal SNR allocation are explicit reproduction choices, because the paper does not disclose them.

Data Generation Pipeline



One sample: QPSK grid → OFDM waveform → EPA multipath fading → noisy received grid → LS pilots + clean channel target.

How One Training Sample Is Generated

1. Transmitted Signal

$$X \in \mathbb{C}^{72\times 14}$$

- Random QPSK symbols.
- Unit-energy constellation.
- 72 subcarriers.
- 14 OFDM symbols.
- CP length = 16.

2. EPA Channel

- 3GPP EPA, 7 paths.
- Carrier = 2.5 GHz.
- Speed = 50 km/h.
- Doppler $\approx$ 115.82 Hz.
- GMEDS fading model.

3. Clean Observation

After channel filtering and delay compensation:

$$Y_{\text{clean}} = H_{\text{TRUE}} \odot X$$

Therefore,

$$\boxed{H_{\text{TRUE}} = \frac{Y_{\text{clean}}}{X}}$$

4. Noisy Observation

$$Y = Y_{\text{clean}} + W$$

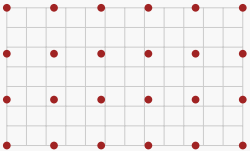
with

$$\sigma_W = \sqrt{\frac{10^{-\text{SNR}/10}}{2}}.$$

LS is then calculated only at pilot positions.

Pilot Configuration

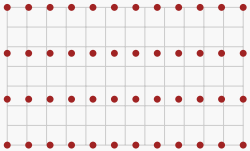
24-Pilot Configuration



$6 \times 4 = 24$

6 frequency positions $\times$ 4 time positions.

48-Pilot Configuration



$12 \times 4 = 48$

12 frequency positions $\times$ 4 time positions.

Important Reproduction Choice

The paper does not specify the exact pilot coordinates. We therefore use uniformly spaced pilot positions and store the coordinates in the dataset metadata.

Paper Replication

AttRNet Replication: Results Across Configurations

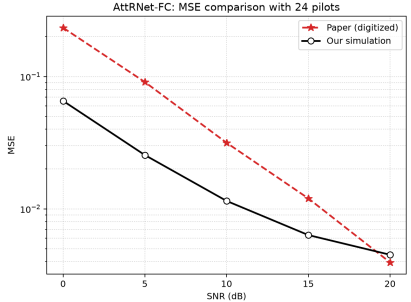
SNR (dB)	24 pilots		48 pilots	
	FC	Conv	FC	Conv
0	0.065501	0.065502	0.038934	0.038917
5	0.025519	0.025524	0.016112	0.016113
10	0.011482	0.011486	0.007570	0.007568
15	0.006338	0.006349	0.004428	0.004420
20	0.004498	0.004511	0.003134	0.003127

- MSE decreases consistently as SNR increases.
- For both architectures, 48 pilots yield lower MSE than 24 pilots.
- FC and Conv variants have very similar results for a fixed pilot count.

Paper Comparison: AttRNet-FC

24 pilots

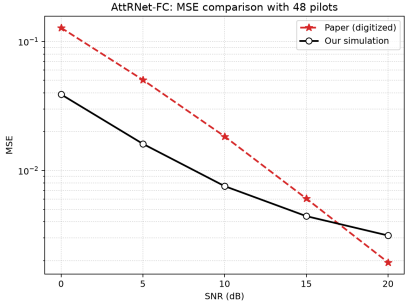
SNR	Paper†	Our MSE	Diff.%
0	0.23400	0.065501	-72.01
5	0.09100	0.025519	-71.96
10	0.03160	0.011482	-63.67
15	0.01200	0.006338	-47.18
20	0.00394	0.004498	+14.16



FC attention (24 pilots): reproduced MSE versus SNR.

48 pilots

SNR	Paper†	Our MSE	Diff.%
0	0.12800	0.038934	-69.58
5	0.05050	0.016112	-68.09
10	0.01840	0.007570	-58.86
15	0.00606	0.004428	-26.93
20	0.00194	0.003134	+61.55

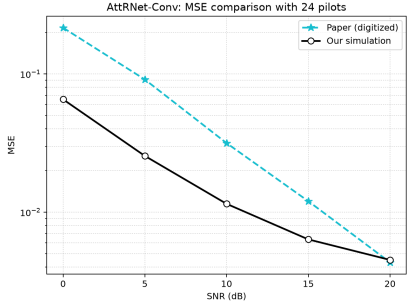


FC attention (48 pilots): reproduced MSE versus SNR.

Paper Comparison: AttRNet-Conv

24 pilots

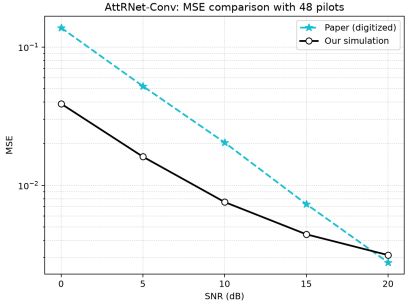
SNR	Paper†	Our MSE	Diff.%
0	0.21600	0.065502	-69.68
5	0.09120	0.025524	-72.01
10	0.03160	0.011486	-63.65
15	0.01200	0.006349	-47.09
20	0.00433	0.004511	+4.18



Conv attention (24 pilots): reproduced MSE versus SNR.

48 pilots

SNR	Paper†	Our MSE	Diff.%
0	0.13800	0.038917	-71.80
5	0.05220	0.016113	-69.13
10	0.02040	0.007568	-62.90
15	0.00731	0.004420	-39.53
20	0.00277	0.003127	+12.89



Conv attention (48 pilots): reproduced MSE versus SNR.

Replication: Interpretation

- The expected decreasing-MSE trend with increasing SNR is reproduced for all four configurations.
- Our values are below digitized paper values at most SNRs, but exceed them at 20 dB for all configurations.
- Since reference values are plot-digitized and the original code/dataset are unavailable, this numerical gap is treated as a replication discrepancy rather than a performance claim.
- Possible sources include pilot coordinates, channel-target definition, noise/SNR convention, normalization, preprocessing, and MATLAB channel settings.

Status

The AttRNet baseline is implemented and all four pilot-count/architecture configurations have been evaluated. The next phase studies pilot overhead and placement.

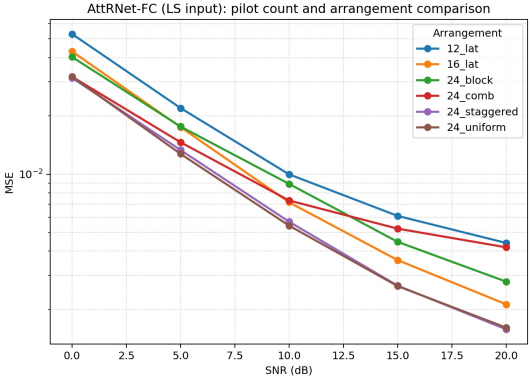
Pilot Reduction: Configurations Tested

- After reproducing AttRNet, we began reducing pilot overhead by testing different pilot counts and arrangements.
- Arrangements evaluated so far include comb, uniform, block, staggered, and lattice-based patterns.
- The goal is to retain channel-estimation accuracy while reducing the number of pilot resource elements.

Comparison protocol

Compare MSE at the same SNR using consistent model settings and evaluation data. Pilot overhead is proportional to the number of pilot resource elements in the grid.

Pilot Reduction: Preliminary Results



Observations

- Uniform placement gives the lowest MSE among the tested 24-pilot arrangements over most SNR values.
- The 16-pilot uniform configuration outperforms the 24-pilot block and comb configurations in the shown results.
- Pilot placement matters in addition to pilot count; more pilots do not automatically guarantee lower MSE if their arrangement is less effective.

Next Steps

Proposed Direction: Residual Transformer Estimator

- **Candidate:** a lightweight Transformer encoder operating on pilot observations and their time-frequency coordinates.
- **Position encoding:** append or embed each pilot's subcarrier and OFDM-symbol indices so self-attention can distinguish pilot locations and learn long-range relationships.
- **Residual reconstruction:** combine the Transformer representation with a lightweight convolutional decoder or upsampling head to reconstruct the full 72×14 channel grid.
- **Why this model:** self-attention can model non-local dependencies across separated pilots, while positional information preserves their locations; a compact design helps control computational cost.
- **Fair comparison:** compare against AttRNet-FC/Conv with matched data splits, pilot patterns, SNRs, and training conditions.

Evaluation criteria

NMSE/MSE, performance versus SNR, pilot overhead, parameter count, FLOPs, and inference time.

Positional encoding is a design hypothesis to test, not a guaranteed improvement. A CNN/Transformer hybrid is a practical first candidate; also retain a small CNN baseline for complexity comparison.

Work Plan Till the End of the Term

Phase	Planned work and deliverable
1. Pilot study	Complete controlled comparisons across pilot counts and comb, uniform, block, and staggered arrangements; report MSE/NMSE versus SNR and pilot overhead.
2. Model design	Implement a lightweight residual Transformer with time-frequency positional encoding and a channel-grid reconstruction head.
3. Benchmarking	Compare with AttRNet-FC/Conv under matched settings using MSE/NMSE, SNR response, pilot overhead, parameter count, FLOPs, and inference time.
4. IRS-assisted OFDM	Adapt the estimator to the IRS-assisted setting, accounting for cascaded-channel estimation and IRS phase configuration.

Expected outcome

A validated low-pilot estimator and a fair, complexity-aware comparison against the AttRNet baseline, followed by an IRS-assisted extension.

Summary and Expected Contribution

- Reproduced AttRNet-FC and AttRNet-Conv and evaluated both with 24 and 48 pilots.
- Confirmed that MSE decreases with SNR and that the 48-pilot setups perform better than the corresponding 24-pilot setups in the current baseline tests.
- Began pilot-overhead reduction; preliminary results favor uniform placement, and 16 uniform pilots outperform the tested 24-pilot block and comb configurations in the supplied plot.
- Next: systematically validate the pilot trade-off, then evaluate a lightweight residual Transformer with time-frequency positional encoding.
- Final direction: investigate IRS-assisted OFDM channel estimation, including cascaded channels and IRS phase configuration.

References I



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